2021. M81A

2021L562A1EL

Coimisiún na Scrúduithe Stáit

State Examinations Commission

Leaving Certificate Examination, 2021

Design & Communication Graphics

Higher Level

Section A (60 marks)

Centre No.

Thursday, 24 June
Morning, 9:30 - 12:30

This examination is divided into three sections:

SECTION A	(Core - Short Questions)
SECTION B	(Core - Long Questions)
SECTION C	(Applied Graphics - Long Questions)

SECTION A

- Four questions are presented.
- Answer **any three** on the A3 sheet overleaf.
- All questions in Section A carry **20 marks** each.

SECTION B and SECTION C

- Eight questions are presented.
- Answer **any two** on drawing paper.
- All questions in Section B and Section C carry **60 marks** each.

General Instructions:

- *Construction lines must be shown on all solutions.*
- *Write the question number distinctly on the answer paper in Sections B and C.*
- *Work on one side of the drawing paper only.*
- *All dimensions are given in metres or millimetres.*
- *Write your Examination number in the box below and on all other sheets used.*

Examination Number

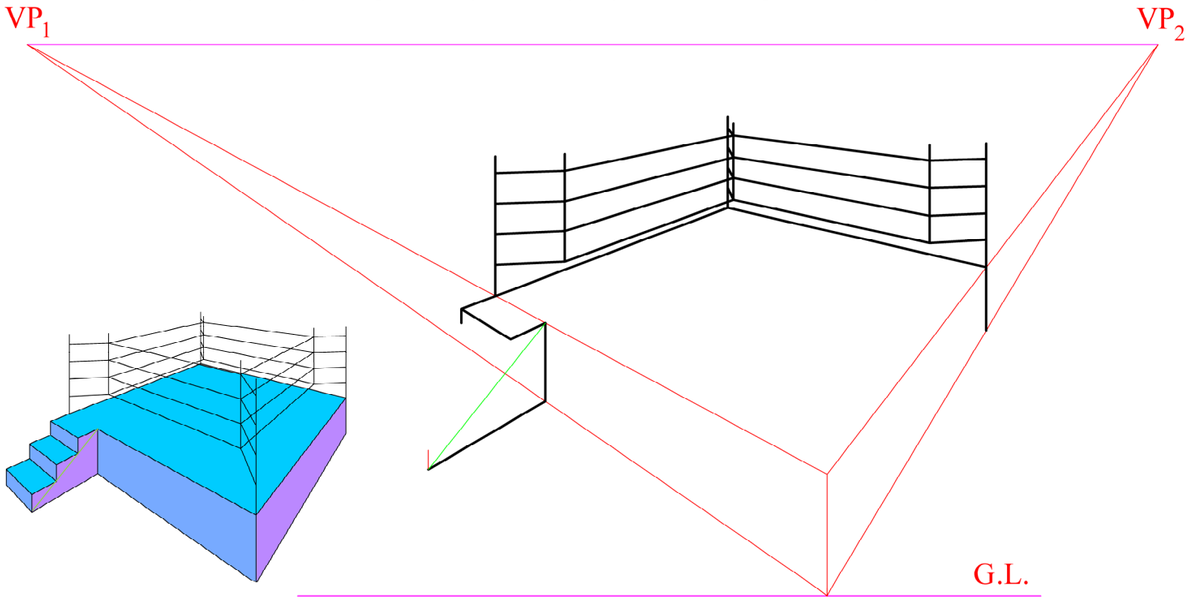
The 2021 examination papers were adjusted to compensate for disruptions to learning due to COVID-19. This examination paper does not necessarily reflect the same structure and format as the examination papers of past or subsequent years.

SECTION A - Core - Answer **any three** of the questions on this A3 sheet.

A-1. The 3D graphic shows a boxing ring with steps. Shown below is an incomplete perspective drawing of the ring and steps. The positions of the ground line and the vanishing points are given.

(a) Complete the perspective drawing of the boxing ring.

(b) The steps are of equal height. Complete the perspective drawing of the steps.

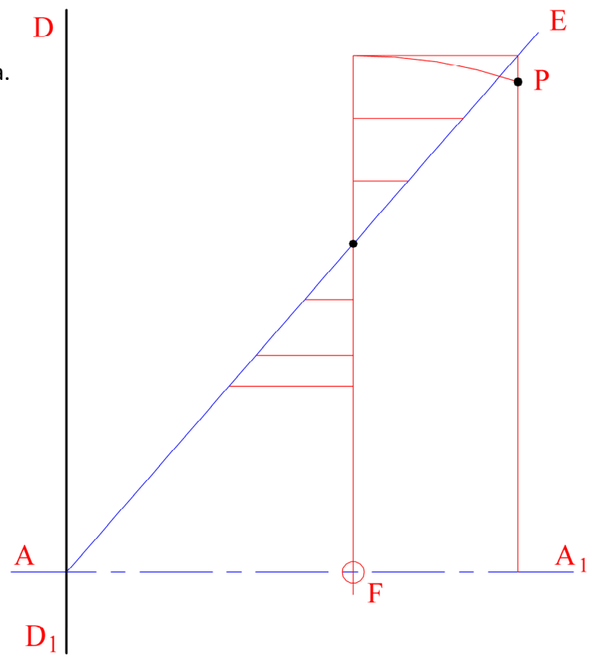


A-2. The image below shows the Spinnaker Tower in Portsmouth. The design includes two hyperbolas. The drawing on the right shows the axis **AA$_1$**, directrix **DD$_1$**, focus **F** and eccentricity line **E** of a similar hyperbola. Two points on the curve are given.

(a) Locate the vertex, five additional points on the curve, and draw the top portion of the hyperbola.

(b) Draw a tangent to the curve at point **P**.

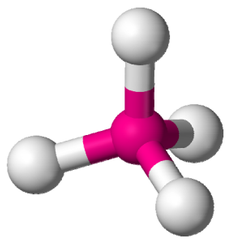


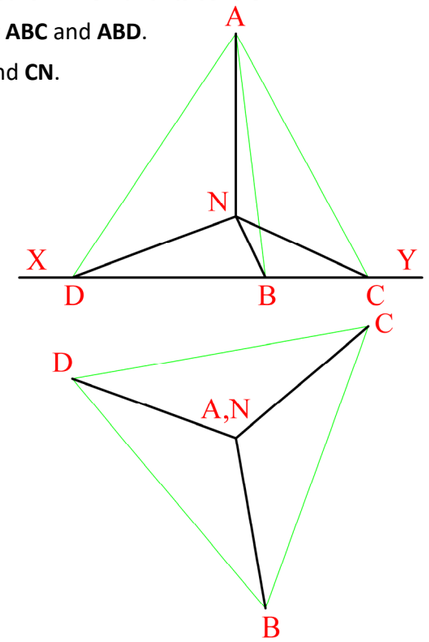


A-3. The image below shows an ammonium ion which has the shape of a regular tetrahedron. A Nitrogen atom is located at the centre of the tetrahedron and four Hydrogen atoms are located at the vertices.
The drawing shows the projections of a regular tetrahedron **ABCD** and its centre **N**.

(a) Determine the dihedral angle between the planes **ABC** and **ABD**.

(b) Determine the true angle between the lines **BN** and **CN**.





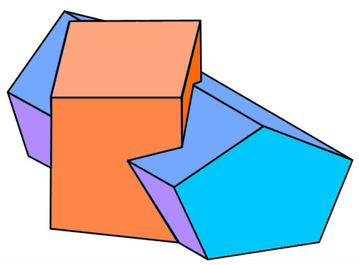
A-4. The image below shows a modern house design which is based on intersecting prisms.

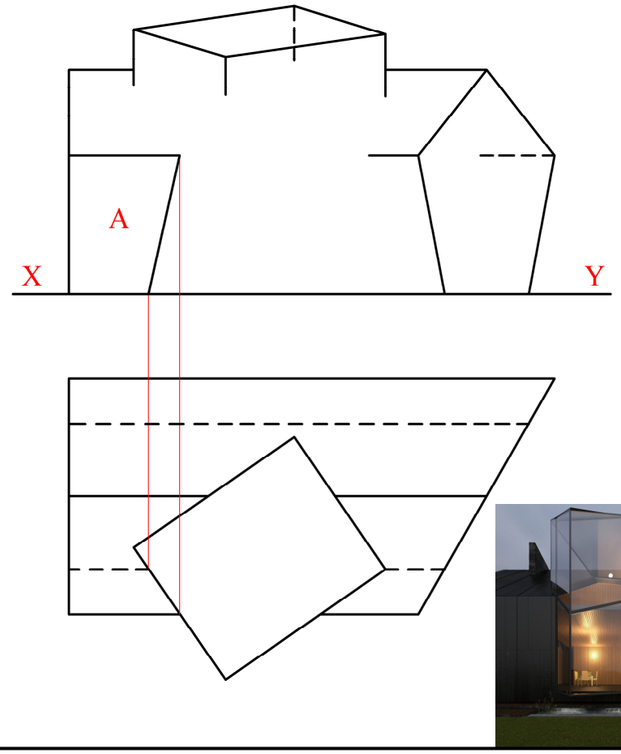
The drawing shows the plan and incomplete elevation of a similar structure which includes a pentagonal prism intersected by a truncated rectangular prism.

A pictorial view of the intersecting solids is given.

(a) Complete the elevation, showing all lines of interpenetration.

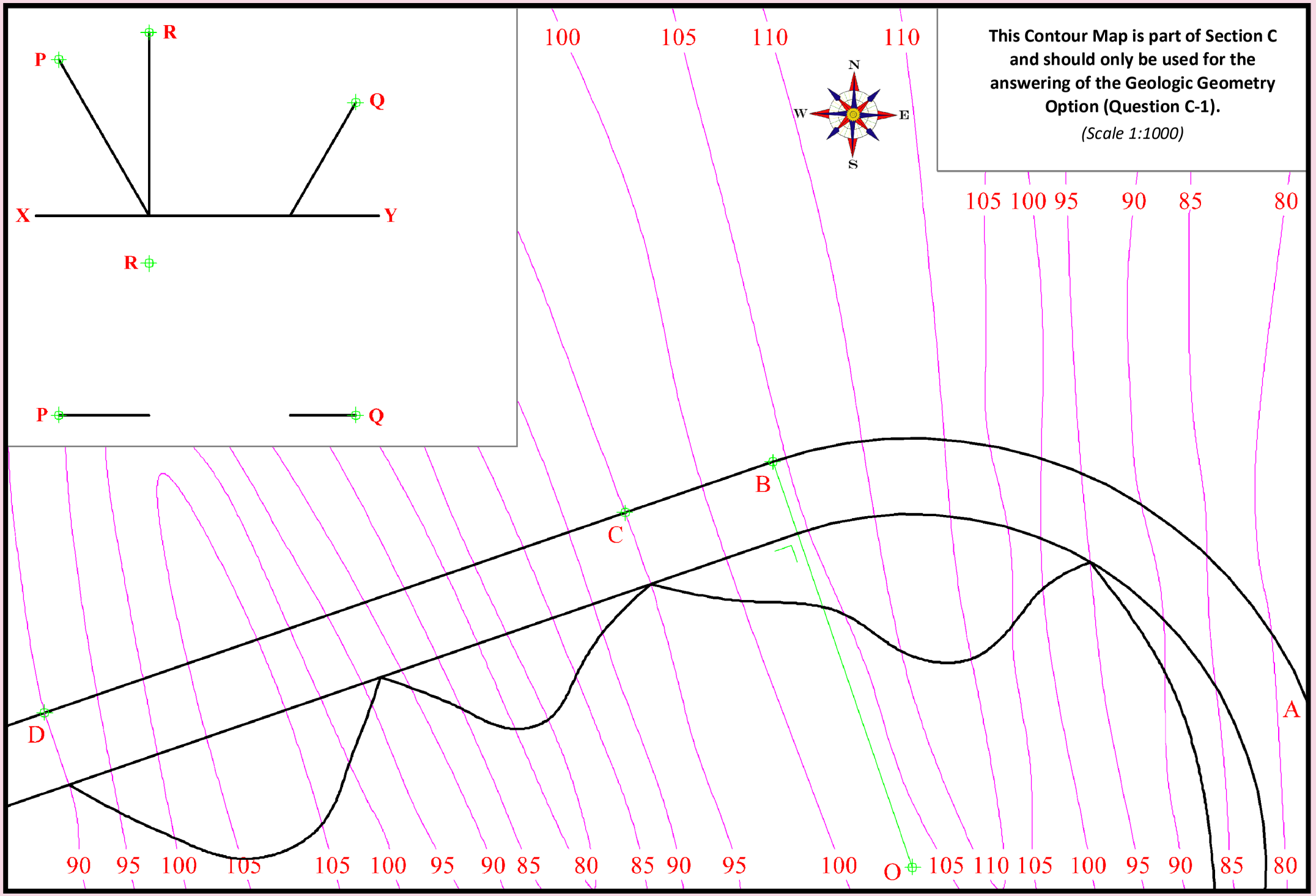
(b) Determine the true shape of the surface **A**.







This examination paper must be returned at the end of the examination – You must include your Examination Number on the front cover.


R
P
Q
X
Y
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100
105
110
110
N
W
E
S
This Contour Map is part of Section C and should only be used for the answering of the Geologic Geometry Option (Question C-1).
(Scale 1:1000)
105 100 95 90 85 80
B
C
A
D
O
90 95 100 105 105 100 95 90 85 80 85 90 95 100 105 110 105 100 95 90 85 80